Sahaj Bhadja

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Summary

Software engineer with 2+ years building AI-powered products, real-time services, scalable APIs, and data-intensive applications using Python, React, TypeScript, cloud infrastructure, recommendation systems, automated testing, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant AI services on Azure SQL and Blob Storage for 500+ users, optimizing schemas, access patterns, and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workers for 1,000+ monthly jobs with explicit states, retries, concurrency controls, and failure isolation, tripling throughput while improving production reliability and operational visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows handling 10,000+ daily requests, integrating secure file ingestion, automated evaluation, and model routing while maintaining stable performance under peak production load.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading and fault-tolerant execution, reducing turnaround by 60% while improving API scalability, testing, and debugging.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release quality while validating integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a customer-facing MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex requirements into maintainable real-time product workflows.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, and defect resolution across frequent customer-facing product updates.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Gathered client requirements and built Flask APIs for booking, scheduling, payment status, and user workflows across 50,000+ records, presenting system architecture and 12+ integration points before implementation.
- Integrated Firebase synchronization, JSON webhooks, and automated payment workflows, reducing manual processing by 70% while improving conversion metric visibility, troubleshooting, and stakeholder communication around recurring production issues.

Skills

Languages & Fundamentals: Python, Java, C++, JavaScript, TypeScript, SQL, Bash, data structures, algorithms, object-oriented design, system design, concurrency, multithreading; **Product & Data:** React.js, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, microservices, PostgreSQL, MySQL, Azure SQL, MongoDB, ETL, real-time processing, data analysis, experimentation, recommendation systems, personalization; **AI & Cloud:** machine learning, deep learning, CNN, LLM orchestration, prompt engineering, AWS, Azure, GCP, S3, Docker, Kubernetes; **Quality & Scale:** Jenkins, Git, CI/CD, PyTest, Mockito, automated testing, debugging, monitoring, performance optimization, fault tolerance, production reliability, Agile/Scrum, product ownership, cross-functional collaboration

Projects

Trend Tracker Live | *Python, CNN, React.js, AWS EC2, Strapi*
- Developed an ecommerce recommendation platform serving 3M+ users, improving personalization accuracy from 78% to 92% through CNN-based modeling while deploying responsive experiences and scalable AWS-backed production services.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a scalable platform for 50,000+ users and 1M+ transactions, combining secure APIs, real-time synchronization, advanced filtering, and Kubernetes-backed deployments for reliable high-volume customer ecommerce-style workflows.

PiXMod Live | *Next.js, TypeScript, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS platform processing 1,000+ assets with semantic search, subscription billing, scalable MongoDB storage, and responsive customer product experiences, increasing premium conversion by 18%.